

Ansys Fluent Simulation Report

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System Information

Application	Fluent
Settings	3d, double precision, pressure-based, laminar
Version	24.1.0-10184
Source Revision	5b3f9fb3c8
Build Time	Nov 22 2023 10:32:41 EST
CPU	11th Gen Intel(R) Core(TM) i7-1185G7 @ 3.00GHz
OS	Windows

Geometry and Mesh

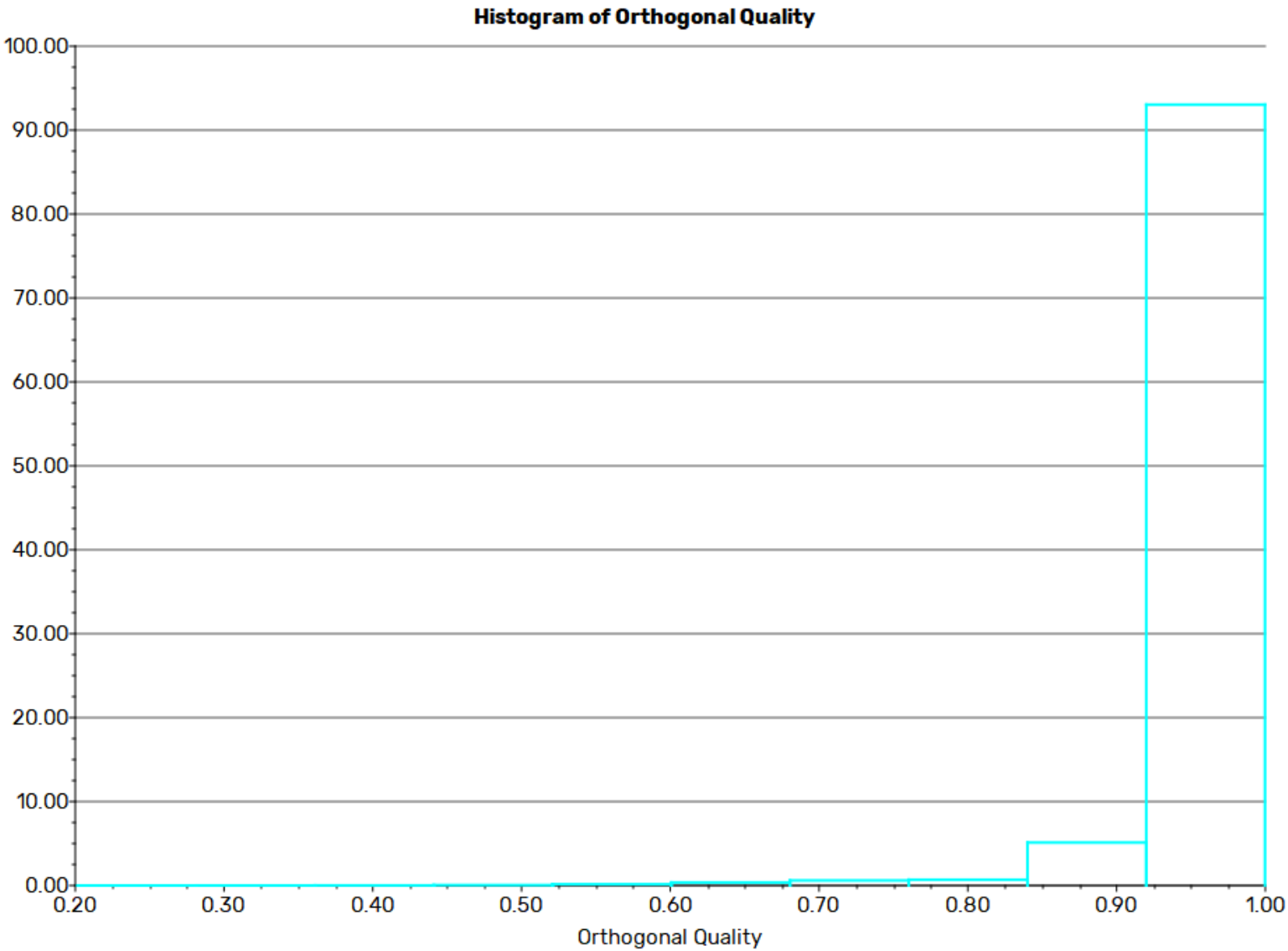
Mesh Size

Cells	Faces	Nodes
169116	873871	592483

Mesh Quality

Name	Type	Min Orthogonal Quality	Max Aspect Ratio
micro-channel-flow-solid	Poly Cell	0.20157245	12.249739

Orthogonal Quality



Simulation Setup

Physics

Models

Model	Settings
Space	3D
Time	Steady
Viscous	Laminar

Material Properties

— Fluid	
— air	
Density	997 kg/m^3
Cp (Specific Heat)	1006.43 J/(kg K)
Thermal Conductivity	0.0242 W/(m K)
Viscosity	0.000855 kg/(m s)
Molecular Weight	28.966 kg/kmol
— Solid	
— aluminum	
Density	2719 kg/m^3
Cp (Specific Heat)	871 J/(kg K)
Thermal Conductivity	202.4 W/(m K)

Cell Zone Conditions

— Fluid	
— micro-channel-flow-solid	
Material Name	air
Specify source terms?	no
Specify fixed values?	no
Frame Motion?	no
Porous zone?	no
3D Fan Zone?	no

Boundary Conditions

— Inlet	
— inlet	
Velocity Specification Method	Magnitude, Normal to Boundary
Reference Frame	Absolute
Velocity Magnitude [m/s]	5e-06

Supersonic/Initial Gauge Pressure [Pa]	0
— Outlet	
— outlet	
Backflow Reference Frame	Absolute
Gauge Pressure [Pa]	0
Pressure Profile Multiplier	1
Backflow Direction Specification Method	Normal to Boundary
Backflow Pressure Specification	Total Pressure
Build artificial walls to prevent reverse flow?	no
Radial Equilibrium Pressure Distribution	no
Average Pressure Specification?	no
Specify targeted mass flow rate	no
— Wall	
— wall-inner	
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip
— wall-outer	
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip
— wall_pillar	
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip

Reference Values

Area	1 m^2
Density	1.225 kg/m^3
Enthalpy	0 J/kg
Length	1 m
Pressure	0 Pa
Temperature	288.16 K
Velocity	1 m/s
Viscosity	1.7894e-05 kg/(m s)
Ratio of Specific Heats	1.4
Yplus for Heat Tran. Coef.	300
Reference Zone	micro-channel-flow-solid

Solver Settings

— Equations	
Flow	True
— Numerics	
Absolute Velocity Formulation	True
— Pseudo Time Explicit Relaxation Factors	
Density	1

Body Forces	1
Explicit Momentum	0.5
Explicit Pressure	0.5
— Pressure-Velocity Coupling	
Type	Coupled
Pseudo Time Method (Global Time Step)	True
— Discretization Scheme	
Pressure	Second Order
Momentum	Second Order Upwind
— Solution Limits	
Minimum Absolute Pressure [Pa]	1
Maximum Absolute Pressure [Pa]	5e+10
Minimum Static Temperature [K]	1
Maximum Static Temperature [K]	5000

Run Information

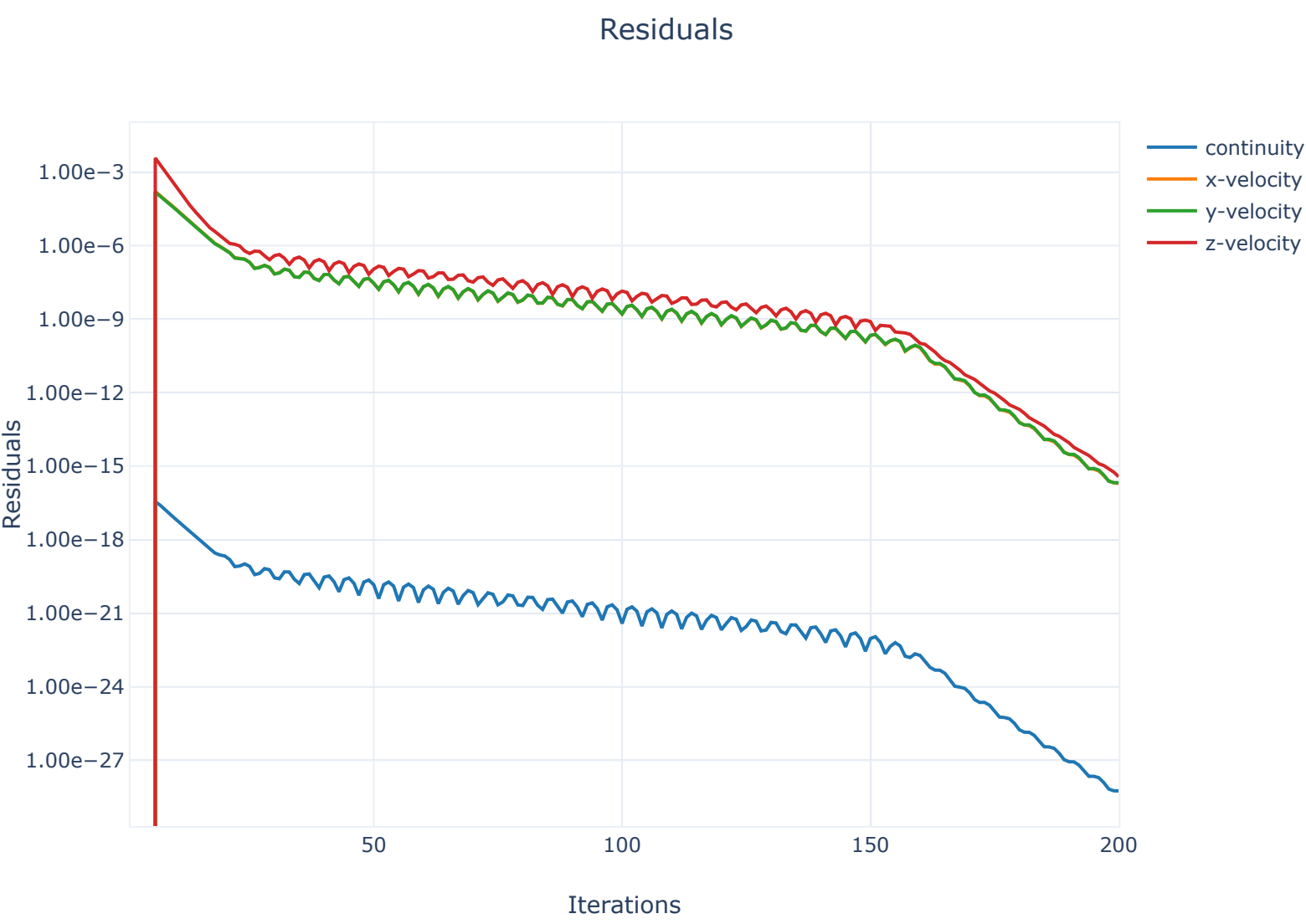
Number of Machines	1
Number of Cores	4
Case Read	11.522 seconds
Data Read	0.616 seconds
Iteration	157.855 seconds
AMG	120.402 seconds
Virtual Current Memory	1.57061 GB
Virtual Peak Memory	2.23973 GB
Memory Per M Cell	7.62089

Solution Status

Iterations: 200

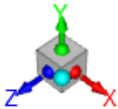
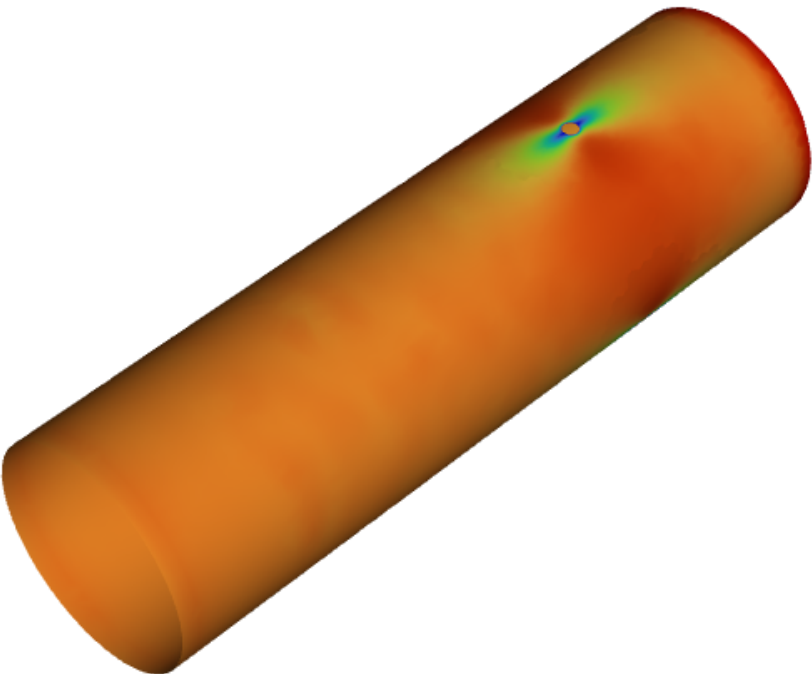
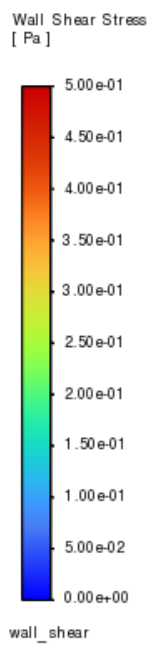
	Value	Absolute Criteria	Convergence Status
continuity	5.640792e-29	0.001	Converged
x-velocity	1.972393e-16	0.001	Converged
y-velocity	2.129027e-16	0.001	Converged
z-velocity	3.750728e-16	0.001	Converged

Residuals

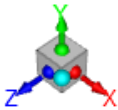
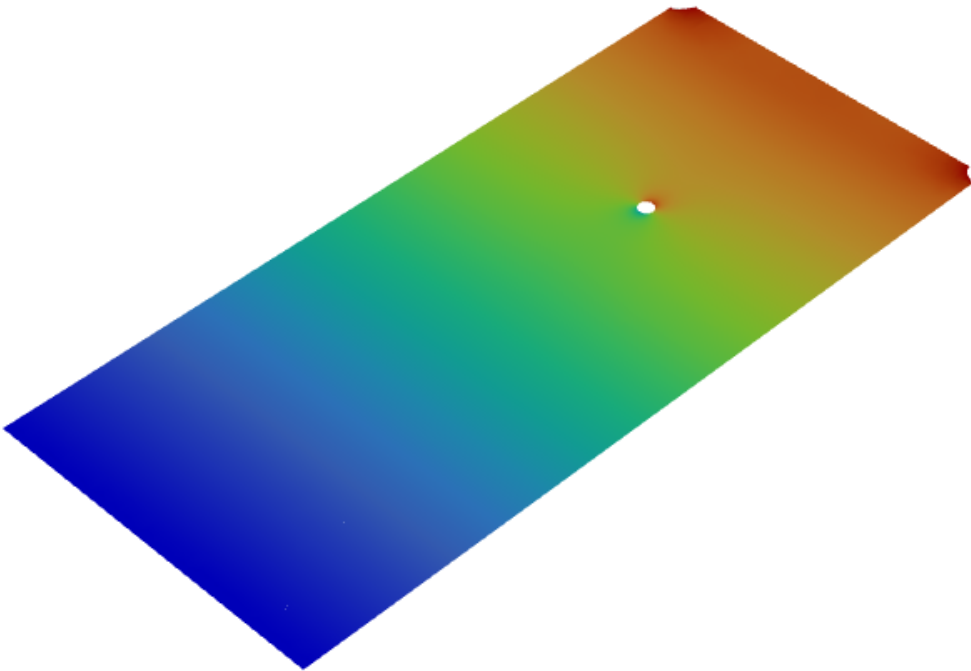
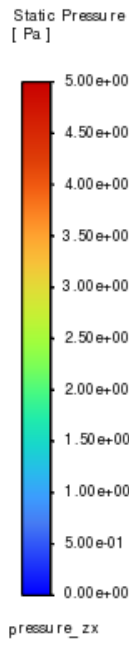


Contours

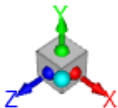
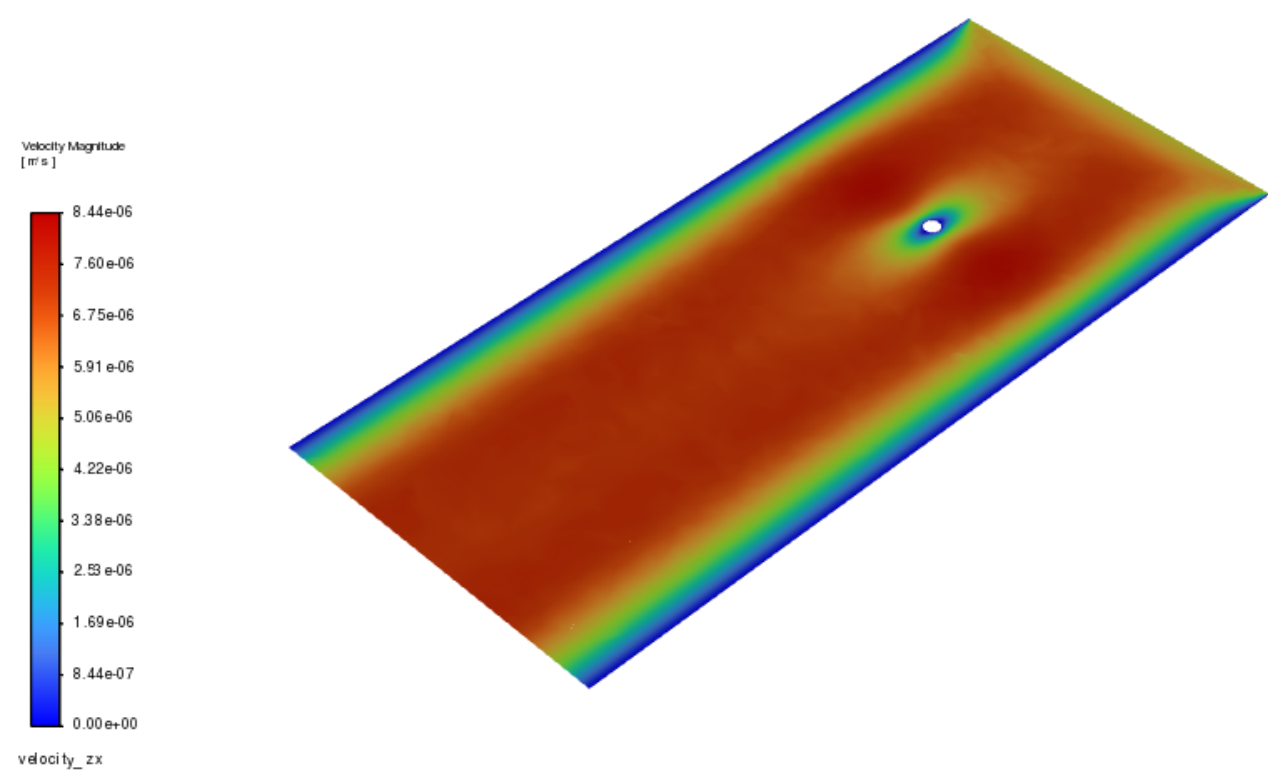
wall_shear



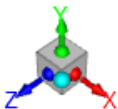
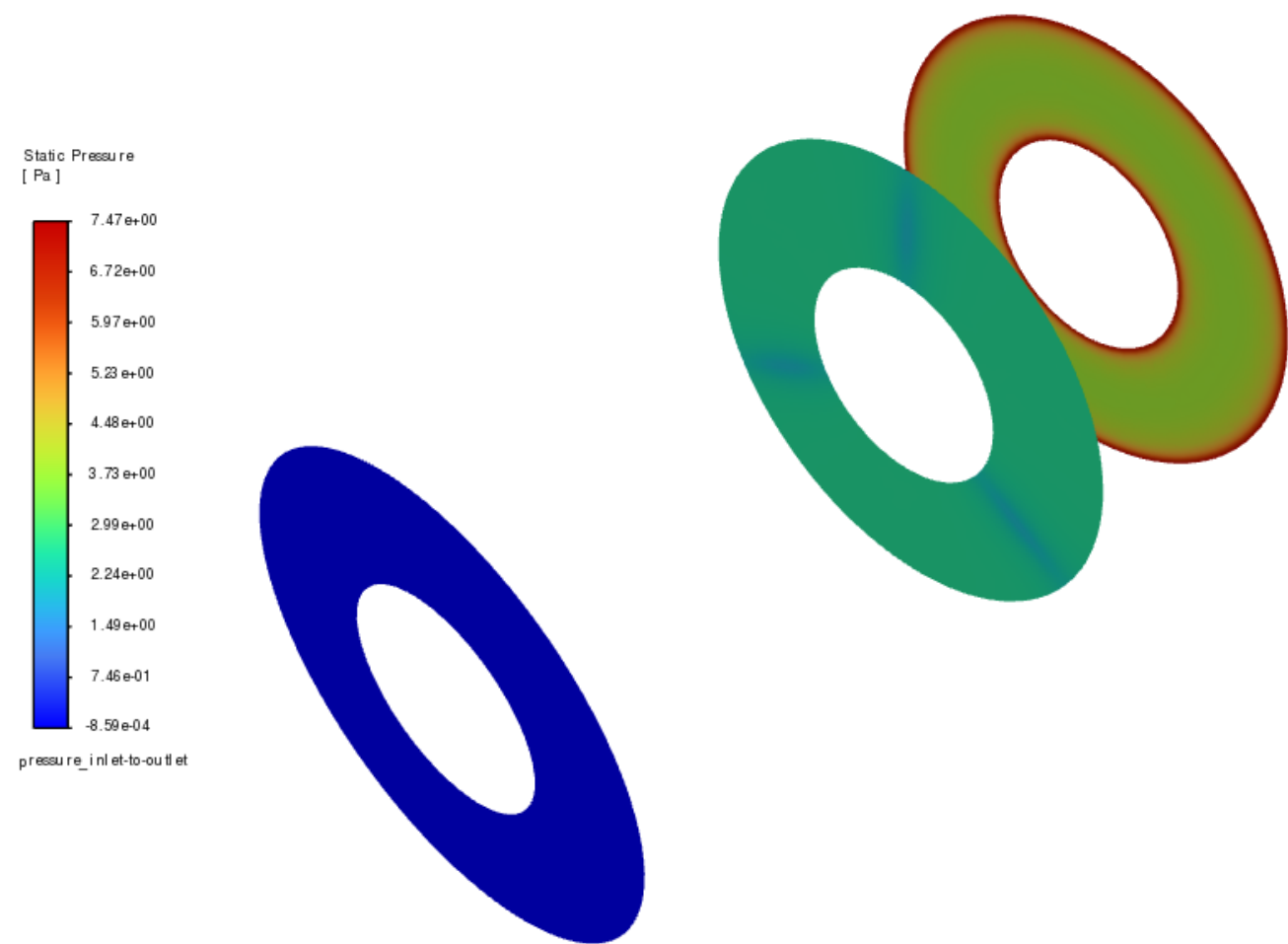
pressure_zx



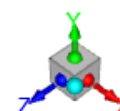
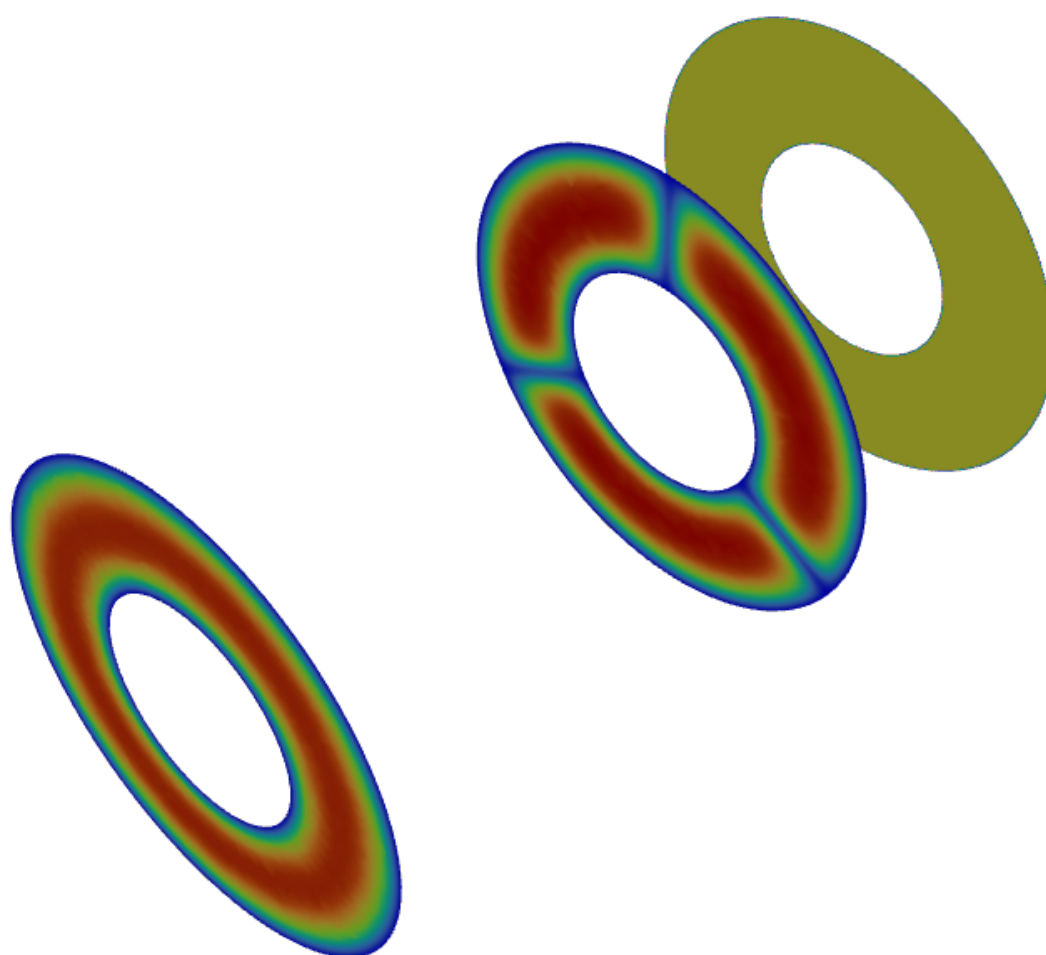
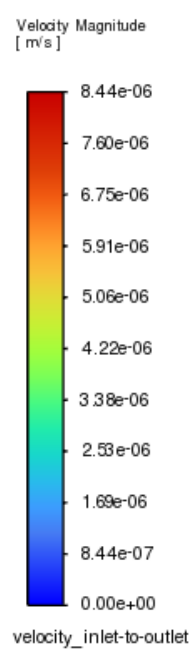
velocity_zx



pressure_inlet-to-outlet



velocity_inlet-to-outlet



Vectors

vectors

